

Document reference		NPS / 002 / 006					
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NPS/002/006 – Technical Specification for Service Cut Outs, Terminal Blocks, Meter Tail Protectors & Pole Mounted Fuse Units.

1. Purpose

This document details CE Electric UK's requirements for LV service termination and associated equipment required for use on the companies distribution network.

This document supersedes the following, all copies of which should be destroyed.

Ref.	Version	Title
NPS/002/006	1 (Nov-05)	Service Cut Outs, Terminal Blocks, Meter Tail Protectors & Pole Mounted Fuse Units.

2. Scope

This document details CE Electric UK's technical requirements for the products detailed below:-

- Heavy duty cut outs for power supplies to premises with loads in excess of 100amps.
- Single and three phase 100 amp house service cut outs and single pole fuse units.
- Insulated terminal blocks for use in conjunction with 100amp cutouts.
- Outdoor pole mounted single-phase fuse units, up to 400amp.
- Meter tail protectors.

3. Heavy Duty Cut Outs

3.1.1 Application

Heavy-duty cutouts shall be suitable for providing power supplies to commercial and industrial premises with loads in excess of 100 amps per phase. Where appropriate cutouts shall be fixed vertically to walls or metal frames in customers' premises. Alternatively they may be installed within compartments in customer's switchboards.

Cutouts shall be capable of providing a PME earth, a SNE earth by means of a removable link and no earth by mean of a blanking plate or insulated cover.

3.1.2 Materials and construction

The cutout enclosure, including the cable termination enclosure, shall be manufactured from insulating material, and shall conform to the relevant requirements of BS7657 relating to impact, insulation / tracking

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and flammability. Flame retardant ABS is a suitable material. Typically, the compartments shall have rating of IP54.

All parts forming the main current path shall be manufactured from hard drawn high conductivity copper or brass, which shall be electro tin plated as required. All ferrous parts must be either stainless steel or adequately plated to prevent corrosion in normal service conditions.

The equipment shall be shielded pattern in accordance with ENATS 37-2 and with an IP2X rating. The permanent shielding shall be readily and safely removable and replaceable, without special tools, to allow adequate access for installing the cables.

The cable termination chamber shall be of a two piece construction. The outer section being removable for cable connections with the inner in-situ. The inner section shall contain captive nuts (or similar) to receive the fixing screw used to joint the two halves. The inner and outer halves of the chamber shall be non-interchangeable.

Cable entries shall be possible from directly below and from either side of the cable chamber. Unused cable entries shall have blanking plugs. Removal of the blanking plugs shall only be possible after entry to the cable enclosure. No part of the cable chamber shall be removable without first removing the fuse compartment cover.

The main fuse compartment shall have a one-piece lid, retained in the closed position by screw fixings. Each of the fixings shall be capable of being sealed in the closed position using standard galvanised sealing wire of Maximum diameter of 2mm.

If wood is used for the top cable entry plate, wall-mounting frame or as cable entry blanking plugs, then this must be a durable hardwood other than oak, be smooth finished and sealed against the ingress of moisture by varnish.

3.2 200 AMP Unit.

3.2.1 Incoming Cable Termination

Range taking shear bolt connectors or mechanical lugs to terminate solid aluminium conductors on 3c CNE 35mm² circular cores and 3c CNE or 4c SNE sector shaped cores up to 95 mm², for cables to BS 7870 -3.40 with stranded copper neutral / earth conductors. Where mechanical lugs are used M12 bolts, nuts, flat washers and spring washers shall be supplied with the cutout.

An additional single piece temporary shield shall be provided. This shield will be fitted over the live incoming terminations, with the main fuses removed, to permit safe working on the outgoing circuit. The shield shall maintain IP2X protection under reasonable mechanical stress arising from the work. It shall be readily and securely fixed without conductive fittings and the use of tools.

3.2.2 Outgoing Cable Termination

Either of two methods is acceptable to terminate the outgoing cables, which could be two cores per phase, ranging between 35mm² and 95mm². If bolted connections in which mechanical lugs (supplied by purchaser) are to be utilised then the unit shall be supplied with M12 bolts, nuts, flat & spring washers. Tunnel terminals are also acceptable for cables up to 2 x 95mm² per phase. The terminals shall be serrated to accept aluminium or copper conductors and shall have two cable pinching screws, which shall be of sufficient size to give adequate purchase to the cable connections and have been satisfactorily load cycle tested with the minimum and maximum conductor sizes recommended.

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3.2.3 Fuse Carrier

The cutouts shall be provided complete with fuse carriers with 82mm fixing centres suitable for BS88-2 type "L" fuses.

The body of the carrier shall be of suitable insulating material, and be so shaped as to allow a definite grip to facilitate the installation or removal of the fuses by hand, while providing an insulated barrier to all live parts.

Fuse carriers shall be of wedge type, and shall be equipped with two thumbscrews for tightening in situ.

The design of these thumbscrews must be such that, when in position, the whole fuse assembly cannot be fortuitously dismantled when the thumbscrews are released to their full extremities. The thumbscrews shall be of insulating material with the same electrical performance as the fuse carrier.

A fuse link **shall not** be supplied with the fuse carrier.

3.2.4 Neutral Terminal

Facility required for connecting upto four neutral cores. Suitable M12 bolts, nuts, flat and spring washers to be supplied with the cutout.

3.2.5 Earth Terminal

The cutout shall be capable of providing a PME earth, a SNE earth by means of a removable link and no earth by mean of a blanking plate or insulated cover.

3.2.6 Dimensions (Approximate)

H = 480mm max, W = 370mm max, D = 210mm max.

3.2.7 Top Plate

Hardwood type divided along its length or closed cell neoprene.

3.3 400 AMP Unit

3.3.1 Incoming Cable Termination

Range taking shear bolt connectors to terminate solid aluminium cores on 3c CNE or 4c SNE sector shaped conductors between 95 mm² and 185mm², for cables to BS 7870 -3.40 with stranded copper neutral / earth conductors. An additional single piece temporary shield shall be provided. This shield will be fitted over the live incoming terminations, with the main fuses removed, to permit safe working on the outgoing circuit. The shield shall maintain IP2X protection under reasonable mechanical stress arising from the work. It shall be readily and securely fixed without conductive fittings and the use of tools.

3.3.2 Outgoing Cable Terminals

PVC insulation and sheath, stranded copper or aluminium conductors in the range 95 mm² - 185mm² will be terminated using compression or mechanical lugs. Suitable M12 bolts, nuts, flat washers and spring washers shall be provided.

A clear distance of 105mm shall be provided between the centre of the fixing hole and the top plate. It shall be possible to connect two outgoing cores per terminal. Back to back connection is acceptable.

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3.3.3 Fuse Carrier

The cutouts shall be provided complete with fuse carriers with 82mm fixing centres suitable for BS88-2 type "L" fuses.

The body of the carrier shall be of suitable insulating material, and be so shaped as to allow a definite grip to facilitate the installation or removal of the fuses by hand, while providing an insulated barrier to all live parts.

Fuse carriers shall be of wedge type, and shall be equipped with two thumbscrews for tightening in situ.

The design of these thumbscrews must be such that, when in position, the whole fuse assembly cannot be fortuitously dismantled when the thumbscrews are released to their full extremities. The thumbscrews shall be of insulating material with the same electrical performance as the fuse carrier

A fuse link **shall not** be supplied with the fuse carrier.

3.3.4 Neutral Terminal

Facility for connecting upto four neutral cores. Suitable M12 bolts, nuts, flat and spring washers to be supplied with the cutout.

3.3.5 Earth Terminal

The cutout shall be capable of providing PME earth, SNE earth by means of a removable link and no earth by mean of a blanking plate or insulated cover.

3.3.6 Dimension (Approximate)

H = 650 max, W = 470 max, D = 235 max.

3.3.7 Top Plate

Hardwood type divided along its length or closed cell neoprene.

3.4 500 AMP Unit

3.4.1 Incoming Cable Termination

Range taking shear bolt connectors to terminate solid aluminium cores on 3c CNE or 4c SNE sector shaped conductors up to 300mm², for cables to BS 7870 -3.40 with stranded copper neutral / earth conductors An additional single piece temporary shield shall be provided. This shield will be fitted over the live incoming terminations, with the main fuses removed, to permit safe working on the outgoing circuit.

The shield shall maintain IP2X protection under reasonable mechanical stress arising from the work. It shall be readily and securely fixed without conductive fittings and the use of tools

3.4.2 Outgoing Cable Termination

The terminals shall enable stranded or solid copper or aluminium cored cables up to 300mm² to be terminated using mechanical lugs. Suitable M12 or M16 bolts, nuts, flat and spring washers shall be supplied with the cutout.

A clear distance of 105mm shall be provided between the centre of the fixing hole and the top plate. It shall be possible to connect two outgoing cores per terminal. Back to back connection is acceptable.

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3.4.3 Fuse Carrier

The cutouts shall be provided complete with fuse carriers with 92mm fixing centres suitable for BS88-2 type "L" fuses.

The body of the carrier shall be of suitable insulating material, and be so shaped as to allow a definite grip to facilitate the installation or removal of the fuses by hand, while providing an insulated barrier to all live parts.

Fuse carriers shall be of wedge type, and shall be equipped with two thumbscrews for tightening in situ.

The design of these thumbscrews must be such that, when in position, the whole fuse assembly cannot be fortuitously dismantled when the thumbscrews are released to their full extremities. The thumbscrews shall be of insulating material with the same electrical performance as the fuse carrier.

A fuse link **shall not** be supplied with the fuse carrier.

3.4.4 Neutral Terminal

Facility to connect four neutral cores, including one neutralising core. M12 bolts, nuts and washers to be supplied with the cutout.

3.4.5 Earth Terminal

The cutout must be capable of providing a PME earth, a SNE earth by means of a removable link and no earth by mean of a blanking plate or insulated cover

3.4.6 Top Plate

Closed cell neoprene or equivalent.

3.4.7 Dimensions (Approximate)

H = 700 max, W = 500 max, D = 210 max.

3.5 Single Phase & Three Phase 100 AMP House Service Cut Out's

3.5.1 Application

The cutout shall be suitable for providing power supplies to commercial or domestic premises with loads up to 100 amps per phase. Cutouts will be fixed vertically onto a meterboard located internally or alternatively they may be installed within an outdoor meter cabinet on the customer's premises.

3.5.2 Materials and construction

The unit is to be moulded in Glass Reinforced Polyester (GRP) or a similar material.

Access to all incoming and outgoing neutral and earth terminals shall be prevented by means of an interlocking insulated cover; the cover shall not be removable unless the fuse carrier is fully removed.

Access to specific outgoing earthing terminals shall be provided by means of a knockout.

3.5.3 Type A1 Cut Out

Type A1 cutouts shall consist of a single-phase fuse-unit incorporating **separate neutral and earth terminals**, complete with an appropriate fuse carrier, but not equipped with a fuse link. A separate cable termination cover may be included, if required.

In accordance with clause 7.1.10.1.102 of BS 7657, access to the outgoing terminals shall be provided, e.g. by knockouts.

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3.5.4 Type A2 Cut Out

Type A2 cutouts shall consist of a single phase fuse unit incorporating **combined neutral and earth terminals**, complete with an appropriate fuse carrier, but not equipped with a fuse link. A separate cable termination cover may be included, if required.

In accordance with clause 7.1.10.1.102 of BS 7657, access to the outgoing terminals shall be provided, e.g. by knockouts.

3.5.5 Fuse Holder

The combination of a fuse base with its fuse-carrier. Fuse link not included.

3.5.6 Three Phase Cut Out

This shall be of modular construction. It shall comprise of one type "A" cutout and two single pole fuse units/holders. A separate cable termination cover may be included if required.

3.5.7 Busbar Link

An insulated copper busbar which connects the incoming phase terminal assembly of a type "A" cut out with the incoming phase assembly of a single pole fuse holder to provide a two-way outgoing fused single phase supply. See clause 7.1.7.2.101 of BS 7657.

3.5.8 Terminals

Two incoming and outgoing terminals shall be provided on both phase and neutral in accordance with clause 7.1.7.1 of BS 7657 and shall have capacity for cable core up to 35mm² cross section.

In addition, access to outgoing earth terminals shall be provided for the user without breaking seals in accordance with Clause 7.1.11 of BS 7657 for type A1 or A2 fuses. Knockouts are an acceptable means of access.

3.5.9 Fuse Carrier

Fuse carrier shall have a rated continuous current of 100 amp shall fit all associated fuse bases. The fuse carrier shall accept fuse-links type 11b conforming to BS 88-3.

3.5.10 Safety Features

Fixing screws shall not be located directly between phase and neutral blocks such that in the event of heat damage to the insulation, the fixing screw might cause flashover.

Tamper resistant blanking plugs e.g. "golf tee" type, or knockouts, are required to protect unused cable or busbar entry positions.

The provision of seals shall conform to Clause 7.1.11 of BS 7657 and shall meet the mechanical test requirements of Clause 8.2.2.102.

3.5.11 Cable Cover

The cable cover is for the protection of incoming cables. The cover shall be capable of interlocking with the appropriate fuse base so that the cover cannot be removed until the fuse carrier has been withdrawn. The cable cover shall have a minimum of two cable entry ports provided at the bottom for which grommets shall be supplied.

This provision shall also apply to the cable cover of 3 phase cutouts.

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3.5.12 Mounting

Suitable fixing screw(s) e.g. Pozidrive Twinfast, shall be supplied for fixing component parts to a 12mm thick wood chip mounting board. The cable cover of three phase cutouts shall have three screw fixings.

3.6 INSULATED TERMINAL BLOCKS

3.6.1 Materials and construction

The terminal blocks shall be Type C1 as classified in BS7657 and shall generally conform to and be tested in accordance with that specification. They shall be suitable for terminating phase, neutral or earth conductors.

3.6.2 Connector Block

The connector block shall be fabricated from brass and shall be totally enclosed in a moulded insulated housing. The housing cover shall be held in place by a suitably robust screw with sealing tabs.

3.6.3 Seals

The provision of seals shall conform to Clause 7.1.11 of BS 7657 and shall meet the mechanical test requirements of Clause 8.2.2.102.

3.6.4 Terminals

The terminal assembly shall have at least five ways. Each terminal way shall be suitable for terminating aluminium or copper conductors up to 35-mm² cross-section.

3.6.5 Mounting

A suitable fixing screw(s) e.g. Pozidrive Twinfast shall be supplied for fixing to a 12mm thick wood chip board, such that the terminal block is firmly retained or prevented from turning. The fixing screw shall not pass through any conducting part of the connector block.

3.7 OUTDOOR POLE MOUNTED SINGLE PHASE FUSE UNITS

3.7.1 Application

The pole-mounted fuse unit shall be used to provide protection or to allow sectioning or sub fusing of low voltage overhead networks.

The fuse unit shall be capable of being fixed to a wooden pole with a minimum radius of 90mm, by a single point fixing so as to prevent any movement during cable installation and fuse carrier removal and insertion.

3.7.2 Materials and construction

The fuse unit shall be moulded from Glass Reinforced Polyester (GRP) or similar material, and shall comprise of all parts to form a complete device as defined in BS7656.

A Suitable hot-dipped galvanised M12 coach screw in accordance with BS EN 1461 shall be provided for fixing the fuse unit to a wood pole

3.7.3 Fuse Carrier

Fuse carriers shall be suitable for accepting fuse-links, which conform to the general requirements of BS88 parts 1 and 5, with fixing centres of 82mm.

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The body of the carrier shall be of suitable insulating material, and be so shaped as to allow a definite grip to facilitate the installation or removal of the fuses by hand, while providing an insulated barrier to all live parts.

The fuse carrier shall be equipped with two thumbscrews for tightening in situ. The design of these thumbscrews must be such that, when in position, the whole fuse assembly cannot be fortuitously dismantled when the thumbscrews are released to their full extremities.

3.7.4 Current Rating

The fuse unit shall have a continuous current rating of 400-AMPS.

3.7.5 Terminations

They shall enable cables with aluminium or copper conductor up to and including 185-mm² to be terminated using compression lugs. One M12 stud or bolt supplied with one plain and one spring washer and full nut shall be provided for each terminal.

3.8 METER TAIL PROTECTORS: Encapsulated Connector Block

3.8.1 Application

The encapsulated connector block is used to provide an electrical connection between a single rate Electricity Meter manufactured to BSEN 61036:1997 and a standard single-phase domestic service cutout.

3.8.2 Materials and construction

The unit shall consist of two conductors partially encased in a solid insulating block with insulated 'Lead ins'. The conductors shall be stranded copper and have a cross-sectional area of 25mm².

The block shall be Grey or Black in colour and be either the integral solid type or the filled box type. In either case it shall consist of insulating material which comply with the Flame Test specified in BS 7657: 1993 Single Phase Domestic Cut-out's up to 100A Rating.

The position and lengths of the conductor tails on the upper (meter) side of the block shall be compatible with the standardised terminals of a BSEN 62053 Electricity Meter.

The connector block, when assembled with a meter and cutout shall be such that access to unused phase terminals in the top of the cutout is prevented.

The connector block shall generally comply with the criteria laid down in the voltage, insulation resistance and temperature rise tests of BS 7657: 1993. The temperature rise test being carried out with the block installed between a BS 7657:1993 cutout and a BSEN 62053 Single-Phase Meter. The temperature rises shall not exceed those given in table 103 of BS 7657: 1993.

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4. References

4.1 External Documentation

The proposed items shall comply with the relevant International standards, British Standard Specifications and all relevant Energy Networks Association Technical Specifications (ENATS) current at the time of tendering, except where varied by this standard. In respect the following documents are particularly relevant.

Reference	Title
BS 7870 -3.40	LV and MV Polymeric Insulated Cables....
BS 7657	Fuses (cut-outs), ancillary terminal blocks and interconnecting units up to 100 A rating, for power supplies to buildings
BS7656	Specification for low-voltage pole mounting fuses (cutouts). 315 A rating
BS 88	Low-voltage Fuses
BSEN 62053	Electricity Metering Equipment (a.c)
BSEN 60529	Specification for the degree of protection provided by enclosures (IP Code)
BSEN 1461	Hot dip galvanized coatings on fabricated iron and steel articles. Specifications and test methods
ENATS 37-2	Substation Cable Distribution Boards

4.2 Internal documentation

Reference	Title
NIL	

5. Definitions

Term	Definition
NIL	

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
V Gorny	Standards Engineer	V Gorny	15/10/09

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
M Nicholson	Head of System Strategy	M Nicholson	04/11/2009

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
R McMahon	Standards Engineer	R McMahon	16/10/09

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	30/11/2009

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	05/11/2009

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	09/11/2009

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Appendix 1 – Requirements

Description	NEDL - YEDL Catalogue Number
Cut Out 100Amp Single Pole with Combined Neutral / Earth	041715
Fuse Unit 100Amp Single Pole.	041723
Single Phase Cable Cover.	049874
Triple-Pole Cable Cover.	049890
Single Extension Bus-Link	046664
Cut -Out 100Amp Single Pole Separate Neutral And Earth	041707
Fuse Unit 100Amp Single Pole Cut-out.	040782
Single Phase Cable Cover. (for above)	049916
100Amp Single Pole & Neutral with Integral Sealing Chamber	040584
100Amp Single-Pole Fuse Unit	040550
Bus-bar Two-way Left Hand	046532
Cut-out 100Amp Triple Pole & Neutral Without PME Link But Including an Integral Earth & Including Sealing Chamber	040592
Insulated Connector Block: Type 2	045831
400-Amp Pole Mounted Fuse Unit	184115
200-Amp Heavy Duty Cut-out	183875
400-Amp Heavy Duty Cut-out	183998
500-Amp Heavy Duty Cut-out	183926

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Appendix 2 – Addendum to Supplier Requirements

To enable CE Electric UK to install the product(s) in accordance with the manufacturer's recommendations the Tenderer should provide full instructions detailing the manufacturer's recommended method of installation of each tendered product with due consideration, where applicable, to the range of application.

Where the product(s) are considered to be suitable for a range of applications then the full scope of application within the range should be provided by the Tenderer for each tendered product.

The tooling costs associated with the handling, installation, inspection, maintenance, repair and decommissioning of the product(s) is crucial to CE Electric UK's assessment of the product viability. To enable CE Electric UK to understand and assess the cost of required tooling the Tenderer should provide full details of the tooling recommended for use by the manufacturer for the purpose of handling, installation, commissioning, inspection, maintenance, repair and decommissioning of each tendered product. Where available the Tenderer should provide indicative prices applicable to the recommended tooling.

All cutouts must have successfully passed all of the tested described in BS7657.

If a special tool is required to affix the unit then this shall be stated at the time of tender.

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Appendix 3 – Logistical Requirements

The Tenderer shall ensure that each item is suitably packaged and protection to maintain the product and packaging as “fit for service” prior to installation taking account of the potential for an outdoor storage environment. All packaging shall be sufficiently durable giving regard to the function, reasonable use and contents of the packaging. Where product packages tendered are made up of sub packages all the sub packages shall unless varied by this specification, be supplied securely packaged together. Where items are provided in bagged/boxed form the material from which the bags are manufactured shall be capable of sustaining the package weight and resisting puncture by the materials within. Tenderer shall submit at the time of tendering the details of the proposed packaging (i.e. materials composition and structure) to be used for each product. Where the Tenderer is unable to provide packaging suitable for outdoor storage then this should be stated at the time of tender.

Palletised goods shall be supplied on standard 1200mm x 1000mm pallets.

Clearly legible, easily identifiable, durable and unambiguous labelling shall be applied to each individual and where relevant multiple package of like products. Where products packages tendered are made up of sub packages each sub packages shall be marked. As a minimum requirement the following shall be included: -

- * Manufacturer's trademark or name
- * Supplier's trademark or name
- * Description of item
- * Date of packaging and/or batch number
- * Purchasing company Brand or name
- * Purchasing company product code
- * Weight

To enable CE Electric UK to store the product(s) in accordance with the manufacturer's recommendations the Tenderer should provide details of the recommended storage environment with respect to each tendered product.

Details should be provided where relevant in respect to the minimum and maximum exposure levels, frequency of exposure and duration of exposure of the packaged item with respect to: -

- | | |
|-----------------------|-------------------------|
| * Ambient Temperature | * Atmospheric Corrosion |
| * Humidity | * Impact |
| * Water | * Vibration |
| * Dust | * Solar Radiation |

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Appendix 4 - SELF CERTIFICATION CONFORMANCE DECLARATION

Service Cut Outs, Terminal Blocks, Meter Tail Protectors & Pole Mounted Fuse Units required to be supplied against this specification shall comply with the latest issues of the relevant ENATS, British and International Standards specified. The following tables are intended to amplify and/or clarify the requirements of elements of these Standards but do not preclude meeting all requirements of the standards. .

The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product

Cs1 = The product conforms fully with the requirements of this clause

Cs2 = The product conforms partially with the requirements of this clause

Cs3 = The product does not conform to the requirements of this clause

Cs4 = The product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary.
- When any other code is entered the reason for non-conformance shall be entered.
- Prefix each remark with the relevant 'BS EN' 'IEC' or 'ENATS' as appropriate.

Manufacturer:

Product Reference:

Details of the

Name:

Signature:

Date:

NOTE: One sheet shall be completed for each type of product offered.

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Heavy Duty Cut Outs				
General	Clause / Requirements	Conformance Code	Remarks / Comments	
Mounting	BS 7657 6.3 (Suitable for Vertical Fixing)			
IP Rating	IP 54			
Manufactured from moisture resistant material conforming to :- Impact Insulating and tracking Flammability				
	BS 7657 Part 8.2.2.101			
	BS 7657 Part 8.3.3.4			
	BS 7657 Part 8.2.1.101			
Screening / Shielding	ENATS 37-2 and IP2X			
Cable Termination Chamber:-	Suitable for 3 or 4c 3Ø Cables to BS 7870-3.40			
	Two Piece (split) Cover / Captive Fixings			
	Bottom Cable Entry			
	Side Cable Entry			
	Cable Entry Blanking Plugs			
Fuse Compartment:-				
Cover	Single Piece			
Cover Fixings	Captive & suitable for sealing with 2mm dia wire			
Fuse Carrier ((without fuse link)	BS88 "L" Type: 82mm centres 200 & 400amp C/O			
	BS88 "L" Type: 92mm centres 500amp C/O			
Temporary Shielding	To prevent inadvertent access to live terminals when fuses removed, IP2X rating			
Outgoing Cable Top Plate	Hardwood type divided along its length or closed cell neoprene			
	Min distance (105mm) between C/L of outgoing cable fixings and top plate (400 & 500amp units)			
Neutral / Earth Terminal	Suitable for 4 x M12 lugs c/w fixings			
	Capable of providing a PME earth connection via blanking plate / cover			
Earth Terminal	Capable of providing a SNE earth connection via blanking plate / cover			

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Heavy Duty CutOuts 200 amp unit:- Incoming Cable Size Terminations Outgoing Circuits Cable Type Size Terminations Typical Dimensions			
		35mm2 to 95mm2	
		Range taking shear bolt connectors or M12 fixing lugs (c/w all fixings)	
		Typically Double Ins 1c Al or Cu Conductor	
		1 or 2 cables / Ø, 35mm2 to 95mm2	
		Range taking tunnel type connectors or M12 lugs (c/w all fixings)	
		H = 480 max, W = 370 max, D = 210mm max	
400 amp unit:- Incoming Cable Size Terminations Outgoing Circuits Cable Type Size Terminations Typical Dimensions			
		95mm2 to 185mm2	
		Range taking shear bolt connectors	
		Typically Double Ins 1c Al or Cu Conductor	
		1 or 2 cables / Ø, up to 185mm2	
		Suitable for M12 lugs (c/w all fixings)	
		H = 650 max, W = 470 max, D = 235mm max.	
500 amp unit:- Incoming Cable Size Terminations Outgoing Circuits Cable Type Size Terminations Typical Dimensions			
		Up to 300mm2	
		Range taking shear bolt connectors	
		Typically Double Ins 1c Al or Cu Conductor	
		1 or 2 cables / Ø, up to 300mm2	
		Suitable for M12 or M16 lugs (c/w all fixings)	
		H = 700 max, W = 500 max, D = 210mm max.	

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Single Phase, Three Phase Service Cut Outs (Type A1 and A2) & Terminal Blocks Units			
General	Clause / Requirements	Conformance Code	Remarks / Comments
Mounting	BS 7657 6.3 (Suitable for Vertical Fixing)		
Material:	Manufactured from GRP or similar		
Impact	BS 7657 Part 8.2.2.101		
Flammability	BS 7657 Part 8.2.1.101		
Cable Termination Cover:	BS 7657 Part 7.1.10.1.101		
Degree of Protection	BS 7657 Part 7.1.11		
Phase / Neutral Cover	BS 7657 Part 7.1.10.1.102		
Earth Terminals ("knockout")	BS 7657 Part 7.1.10.1.102 & 7.1.11 (user access)		
Security Seals	BS 7657 Part 7.1.11 / 8.2.2.102		
Performance Requirements:			
Temperature Rise	BS 7657 Part 7.2.2		
Contact Stability	BS 7657 Part 7.2.2.101		
Mechanical Endurance	BS 7657 Part 7.2.2.102		
Dielectric Properties	BS 7657 Part 7.2.3 & Part 8.3.3.4		
Overload & Short Circuit	BS 7657 Part 7.2.101		
Cut Out Cable Terminals:	Two incoming and outgoing terminals BS 7657 Part 7.1.7 (up to 35mm ² conductors)		
2w Single Phase Supply Facility	Insulated Busbar Link to BS 7657 Part 7.1.7.2.101		
Fuse-holder (without fuse-link)	BS 7657 Part 2.3 (BS 88-1)		
Fuse-carrier (without fuse-link)	BS 7657 Part 7.1.2.101 Rated 100 amps and suitable for fuse-link type 11b to BS 88-3		
3 Ø Cut Outs (additional req):			
Construction Arrangement	Modular, comprising a single type A cut out (fuse-unit and fuse-holder)		
	Additional single pole fuse-holders (x 2)		
	Separate cable termination cover		
Terminal Blocks			
Single Pole	BS 7657 Part 3.106, Type C1		
Size & number of ways	Up to 35mm ² conductors / 5 way		

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Technical Specification for Service Cut Outs, Terminal Blocks, Meter Tail Protectors & Pole Mounted Fuse Units.			
Single Phase Pole Mounted Fuse Units			
General	Clause / Requirements	Conformance Code	Remarks / Comments
Mounting	BS 7656 Part 6 (Suitable for Pole Fixing, min 90mm radius) Single Point Fixing (M12 Coachscrew)		
Material:	BS 7656 Part 7 (GRP or similar)		
Impact	BS 7656 Part 8.2.2.101		
Flammability	BS 7656 Part 8.2.1.101		
Degree of Protection	BS 7656 Part 7.1.11		
Performance Requirements:			
Rating	400 amps (continuous)		
Temperature Rise	BS 7656 Part 7.2.2 (based on 400amp rating)		
Contact Stability	BS 7656 Part 7.2.2.101 (based on 400amp rating)		
Dielectric Properties	BS 7656 Part 7.2.3 & Part 8.3.3.4		
Overload & Short Circuit	BS 7657 Part 7.2.7 (based on 400amp rating)		
Terminals:	BS 7656 Part 7.1.7 (up to 185mm ² conductors)		
Fuse-carrier (without fuse-link)	BS 7656 Part 7.1.2.101 (suitable for fuse-link to BS88-2 Type "L", 82mm fixing centres)		
Meter Tail Protectors : Encapsulated Connector Block			
Integral solid or filled box type unit made from insulated material to:-			
Impact	BS 7657 Part 8.2.2.101		
Insulating and tracking	BS 7657 Part 8.3.3.4		
Flammability	BS 7657 Part 8.2.1.101		
Temperature Rise Test	BS 7657 Table 103		
Colour	Black or Grey		
Security	Prevent access to unused cutout terminals		
Conductors	25mm ² Cu conductors & insulated "lead in's"		
Interface between:	1Ø Meter to BSEN 62053 & cutouts to BS7657		